MILITARY SPECIFICATION

ADAPTION KIT SUBASSEMBLY, ATOMIC WEAPON, TRAINING: M3

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of atomic weapon control assembly entitled as Adaption Kit Subassembly, Atomic Weapon, Training M3, which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48515(PA)	- Accelerometer, Integrating, Atomic Weapon, Training: M7.
MIL-D-48516(PA)	- Dummy Battery Assembly: M70.
MIL-P-48314(PA)	- Plug, Test, Atomic Weapon.
MIL-S-48517(PA)	- Semiconductor Device, Diode: M37
MIL-C-48518(PA)	- Cable Assembly-Switch, Electrical: M15
MIL-T-48519(PA)	- Timer, Sequential: M16 and M30

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedures for.

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DRAWINGS

ORDNANCE CORPS

8800654 - A.K. Subassembly, Atomic Weapon, Training M3.
EL-8800654 - Equipment List.
EL-8884531 - Equipment List.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

PICATINNY ARSENAL

SQAP - 8821926 - Support, Atomic Weapon Control Training.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The control assembly shall comply with all requirements specified on the drawing (dwg) 8800654, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Circuit resistance. With the exception of circuits containing resistors, capacitors or diodes, the circuit resistance between electrically connected pins shall be 0.03 ohm maximum (max) when tested as specified in 4.4.1

3.5 Insulation resistance. The insulation resistance between electrically isolated circuits shall be 20 megohms minimum (min) when tested as specified in 4.4.2.

3.6 Functioning. The control assembly shall exhibit proper functioning charactetistics in all of the 32 test positions as evidenced by a green "GO" indication when tested as specified 4.4.3.

3.7 Pressure retention. The internal pressure drop of the control assembly shall not exceed 2.0 pounds per square inch (psi) after 1 hour of storage when tested as specified in 4.4.4.

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3.8 Environmental testing. The control assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The control assembly shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.8.1 Vibration. The control assembly shall show no evidence of damage or loosening of parts and shall comply with the requirements of 3.4.3.5, 3.6 and 3.7 after being tested as specified in 4.4.5.1.

3.8.2 Humidity. The control assembly shall show no evidence of damage and shall comply with the requirements of 3.5, 3.6 and 3.7 after being tested as specified in 4.4.5.2.

3.9 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.10 Control components. The control components shall have complied with all requirements specified in their respective specifications and SQAP'S listed below and records shall be furnished certifying that the control components have been inspected and accepted in accordance with their respective specifications:

MIL-A-48515(PA)	- Accelerometer, Integrating, Atomic Weapon, Training:M7.
MIL-D-48516(PA)	- Dummy Battery Assembly: M70.
MIL-P-48314(PA)	- Plug, Test, Atomic Weapon.
MIL-S-48517(PA)	- Semiconductor Device Diode: M37.
MIL-C-48518(PA)	- Cable Assembly-Switch, Electrical:M15
MIL-T-48519(PA)	- Timer, Sequential: M16 and M30
SQAP - 8821926	- Support Atomic Weapon Control Training.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8800654) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Vibration (see 3.8.1)	X	
3. Humidity (see 3.8.2)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

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4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristics table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520 1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 A.K. Subassembly, atomic weapon, training M3 less cover (see dwg 8800654).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Harness insecurely laced	Visual
	102. Insulation sleeving missing	Visual
	103. Improper soldering (see 3.9)	Visual
	104. Damaged wiring	Visual
	105. Incorrect location of electrical component	Visual
	106. "O" Ring missing	Visual
	107. Marking missing, incorrect or illegible	Visual
	108. Any component loose or improperly secured	Visual/Manual
	109. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Protective finish damaged to the extent of exposing base metal	Visual
	202. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 A.K. Subassembly, atomic weapon, training M3 (see dwg 8800654).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Marking missing, incorrect or illegible	Visual
	102. Connector pin bent or broken	Visual
	103. Support cracked or damaged	Visual
	104. Cover cracked or damaged	Visual
	105. Component loose or missing	Visual/Manual
	106. Hardware loose or missing	Visual/Manual
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:
201. Protective finish damaged to the extent of exposing base metal.....Visual
202. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation packing and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Functioning	Major
b. Circuit resistance	Major
c. Insulation resistance	Major
d. Pressure retention	Major
e. Vibration	Major

Of these tests, e is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance, insulation resistance functioning and pressure retention. Each control assembly in the lot shall be tested in accordance with 4.4.1 through 4.4.4. If any item fails to comply with 3.4 through 3.7, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. One sample control assembly shall be selected at random from each 25 units or portion thereof produced. The sample shall be tested in accordance with 4.4.5.1. If any item fails to comply with 3.8.1, the lot shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8800654 and 8884531). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The circuit resistance of the control assembly components shall be measured in accordance with test panel. The current shall be limited to five milliamperes.

4.4.2 Insulation resistance. The insulation resistance of the control assembly circuitry shall be measured with an applied potential of 500 ± 50 volts DC in accordance with test panel.

4.4.3 Functioning. The control assembly shall be tested for operational characteristics in accordance with the operating instructions supplied with the T-4087 Atomic Weapon Test Set.

4.4.4 Pressure retention. The control assembly shall be filled with dry air to a pressure of 15 ± 1 psi. After pressurizing, the control assembly shall be allowed to stand for one hour. Upon completion of this one hour period, pressure shall be measured.

4.4.5 Environmental testing (see 3.8).

4.4.5.1 Vibration. The control assembly shall be secured to a vibration platform. A simple harmonic motion having an amplitude of 0.03 inch (0.06 inch max. total excursion) shall be applied, the frequency being varied uniformly between the approximate limits of 10 to 55 Hertz (Hz). The entire frequency range from 10 to 55 to 10 Hz shall be transversed in approximately one minute.

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The motion shall be continuously applied for 30 minutes in each of the three mutually perpendicular axes. Upon completion of this test, the control assembly shall be tested as specified in 4.4.1, 4.4.2, 4.4.3 and 4.4.4.

4.4.5.2 Humidity. The control assembly shall be placed in a humidity test chamber. Within a period of one hour, the chamber temperature shall be raised to 165 ± 2 degrees Fahrenheit (F) and stabilized (see 6.4) at that temperature for two hours. A relative humidity of 95 percent shall be maintained while the dry bulb temperature is 165 ± 2°F. At the end of the two-hour period, the chamber temperature shall be reduced uniformly over a one-hour period to between 68° and 95°F. The relative humidity during the period of decreasing temperature shall be 100 percent. The relative humidity during the period of decreasing temperature shall be 100 percent. The control assembly shall be subjected to a total of five such humidity cycles (one hour of increasing temperature of 95 percent relative humidity, two hours of temperature stabilization at 95 percent relative humidity). The water used in controlling the humidity shall be distilled with a pH value between 6.5 and 7.5 at 77°F. Upon completion of the fifth cycle, the control assembly shall be removed from the test chamber and excessive moisture shall be removed with dry, fitered, compressed air. The control assembly shall then be tested as specified in 4.4.2, 4.4.3 and 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The control assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169 (MU).

5.1.2 Level C. The control assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The control assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

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b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1802 Revision 1 dated 14 December 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A153

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